Preface

Non-relativistic field theory at finite temperature and finite density was invented in the late 1950s for the theoretical description of condensed matter and nuclear matter under standard laboratory conditions. Although it does not involve concepts beyond the Schrödinger equation and statistical mechanics, it offers a most convenient theoretical framework with which to deal with a large number of particles. It is often referred to as the 'Many-Body Problem' or as the 'N-Body Problem'. Relativistic field theory at finite temperature was first studied by Fradkin (1965) and rediscovered ten years later, the main motivation at the time being a description of the phase transition which occurs in the electroweak theory, at a temperature of the order of 200 MeV. This transition is of course of great interest for the history of the early Universe.

In the early 1980s, lattice gauge theories suggested the existence of a deconfined phase of quarks and gluons, which has been called the 'quark–gluon plasma phase', above a temperature which is estimated nowadays to lie around 150 MeV. The possibility of observing this new state of matter in ultrarelativistic heavy ion collisions gave a further boost to the study of finite temperature field theory. Furthermore, Braaten and Pisarski (1990a,b) made a theoretical breakthrough, which enabled them to reorganize perturbation theory, whose original formulation had led to nonsensical results in some circumstances.

It seems appropriate at present to write a review which gives a self-contained introduction to the subject and, at the same time, an account of the recent developments. Needless to say, all the problems are far from being solved, and much remains to be done, for example, in our understanding of some delicate infrared problems and of non-equilibrium phenomena.

After an introductory chapter which is aimed to outline the physical motivations (and which can be skipped at first reading), the book is

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divided into two parts. The first part, chapters 2–5, is devoted to setting up the formalism. Chapter 2 examines the one-dimensional motion of a quantum particle in a potential, in thermal equilibrium: since many important concepts can be understood on this elementary example, it is used in order to introduce path integrals and thermal Green's functions. Chapter 3 describes perturbation theory and Feynman rules in the case of scalar fields, while chapter 4 deals with some simple applications of these rules. Chapter 5 generalizes the results to the case of Dirac and gauge fields. The second part, chapters 6–10, is devoted to the theoretical developments which have occurred in recent years (although some results were known long ago). Chapter 6 introduces hard thermal loops which are used to describe collective excitations of gauge bosons and fermions. The full perturbative scheme relying on the hard thermal loop concept is developed in chapter 7, which also gives some applications to physical processes. Further applications are found in chapter 8 (dynamical screening) and in chapter 9 (neutrino emission from stars), while chapter 10 concludes with some open problems.

The reader is assumed to have a working knowledge of elementary quantum statistical mechanics, for example at the level of Mandl (1988), and to be familiar with the basic methods of quantum field theory. However, no advanced methods will be needed, other than in chapter 5, where some use is made of the technical details of the quantization of non-Abelian gauge fields in a section which can be skipped at first reading; in particular, renormalization problems will play only a minor role.

In order to keep the book to a reasonable size, and also because of my limited competence, a number of interesting topics had to be omitted. Fortunately I can refer the interested reader to excellent reviews: thermofield dynamics is described in Umezawa, Matsumoto and Tachiki (1982), Landsman and van Weert (1987), van Weert (1994) or Umezawa (1994). Higher order calculations of the partition function in QED and QCD may be found in Kapusta (1989); see Corianò and Parwani (1994) for recent results. References to the literature on symmetry restoration in the electroweak theory can be found in Kapusta (1989) and in Arnold and Espinosa (1993). I have tried to avoid, as far as is possible, overlap with the material already covered by Kapusta (1989), and I decided to leave aside for the time being the whole field of non-equilibrium phenomena in the quark–gluon plasma, which has not yet reached a sufficiently mature stage such that it can be reasonably included in a textbook.

There is an abundant literature on thermal field theory, and I had to make a choice from available references. As a general rule, and except for some historical landmarks, I have not quoted articles prior to 1988, since

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a complete list of references can be found in Landsman and van Weert (1987) and in Kapusta (1989).

While writing this book, I was assisted by many people who I would like to thank warmly. First of all the book would not have been written without the constant encouragement of Peter Landshoff. Jean-Louis Meunier helped me to find my way through the many versions of TEX, Frithjof Karsch provided me with his latest results, given in figures 1.2 and 1.3. Jean-Paul Blaizot, Thierry Grandou, Edmond Iancu, Peter Landshoff and Art Weldon made useful comments on specific points. I am especially indebted to Patrick Reynaud, who took care of the diagrams, and to Rolf Baier, Françoise Guérin, Hubert Mabilat and Toni Rebhan who read a first version of the manuscript thoroughly and made a number of observations and suggestions. The responsibility for the final text is of course entirely mine. Last but not least, Joanna provided me with her moral support.

I cannot conclude this preface without a sad thought for my friend and colleague Tanguy Altherr, whose life ended in a tragic climbing accident at the age of 31. The numerous references to his work bear witness to the importance of his contribution to finite temperature field theory.

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